



Macro Forecasting with the TFT and Fed Speeches

Minna Heim, Chris Truill and Anna Zeitz



Motivation

- Macro forecasts matter for
 - Governments
 - Financial markets
 - Central banks

Motivation

- Macro forecasts matter for
 - Governments
 - Financial markets
 - **Central banks**

Motivation

- Macro forecasts matter for
 - Governments
 - Financial markets
 - **Central banks**
- The Federal Reserve (Fed) has a **dual mandate**:
 1. Price stability
 2. Maximum employment

Motivation

- Macro forecasts matter for
 - Governments
 - Financial markets
 - **Central banks**
- The Federal Reserve (Fed) has a **dual mandate**:
 1. Price stability
 2. Maximum employment
- Accurate forecasts of inflation, unemployment, and GDP are crucial for interest rate decisions

Can we do better than standard Macro Models?

Research Question

- The Fed communicates with the public through official statements and speeches

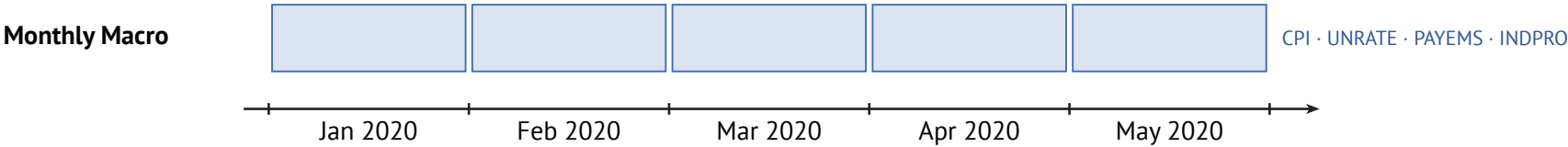
Research Question

- The Fed communicates with the public through official statements and speeches
- It may communicate additional information:
 1. **Superior information**: private data and internal models not available to the public
 2. **Forward guidance**: signals about future policy to guide market expectations

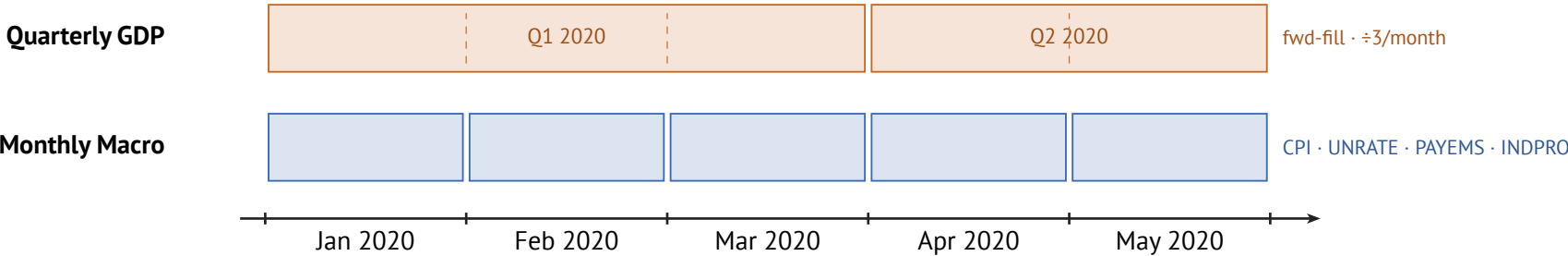
Research Question

- The Fed communicates with the public through official statements and speeches
 - It may communicate additional information:
 1. **Superior information**: private data and internal models not available to the public
 2. **Forward guidance**: signals about future policy to guide market expectations
- ⇒ **Do Fed speeches contain information useful to forecast macroeconomic indicators?**

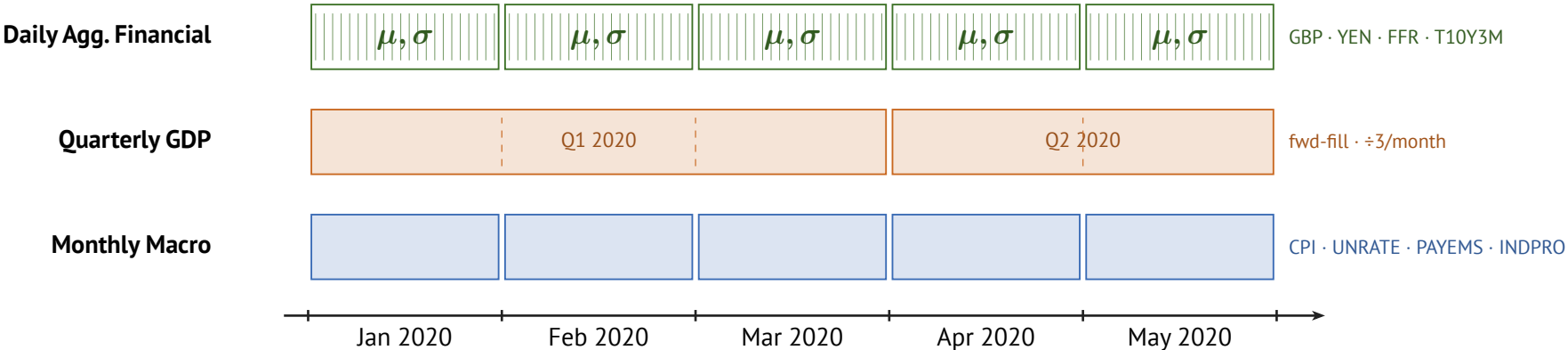
Data Alignment



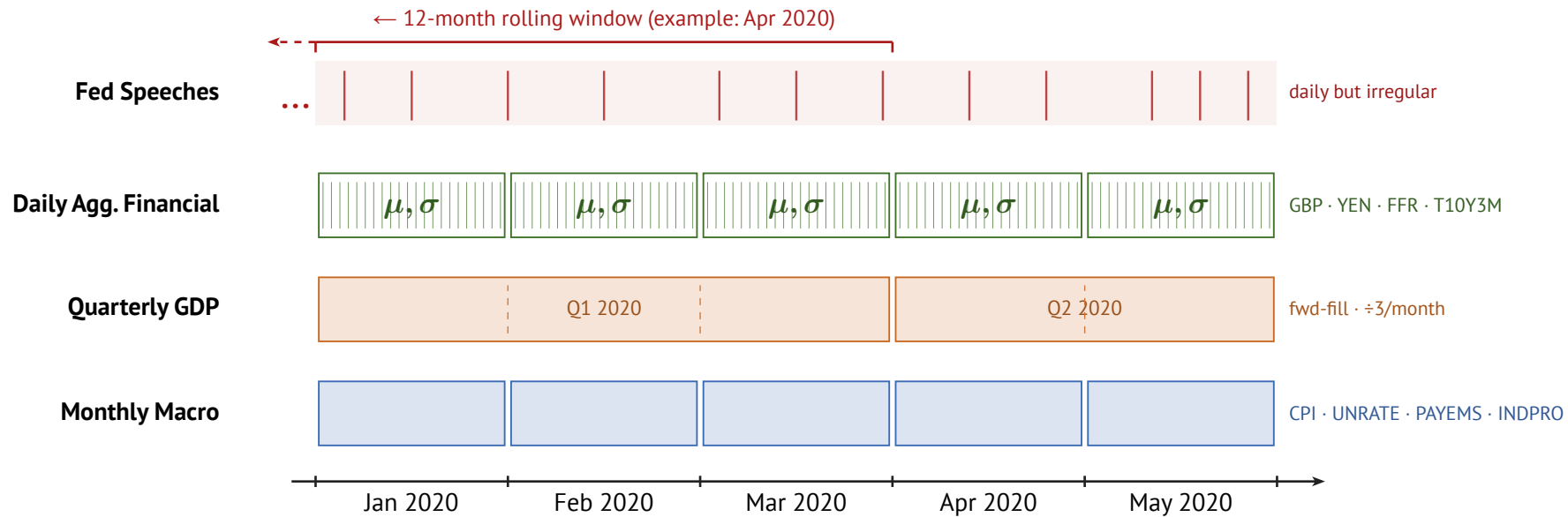
Data Alignment



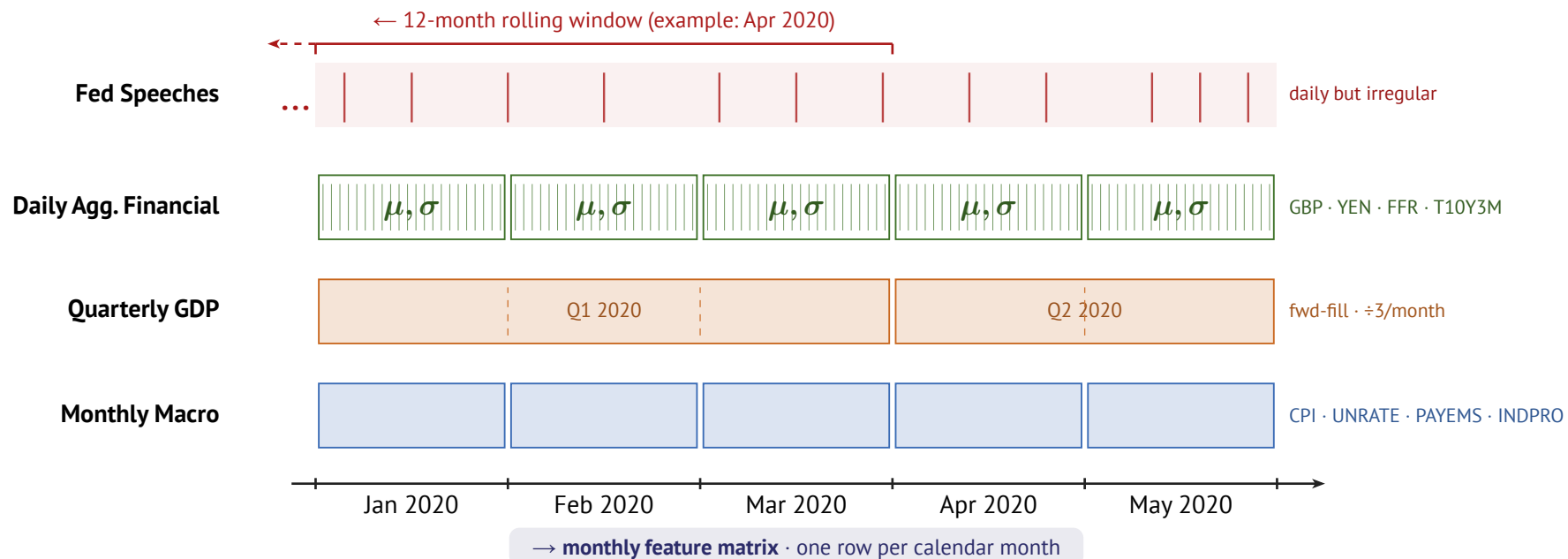
Data Alignment



Data Alignment



Data Alignment



Models

Benchmarks

AR(p)

$$y_t = c + \sum_{i=1}^p \varphi_i y_{t-i} + \varepsilon_t$$

ARIMA(p, d, q)

$$\Delta^d y_t = c + \sum_{i=1}^p \varphi_i \Delta^d y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t$$

■ Integration: d -th difference of y_t ■ MA component: q lags of residuals ε_t

Temporal Fusion Transformer

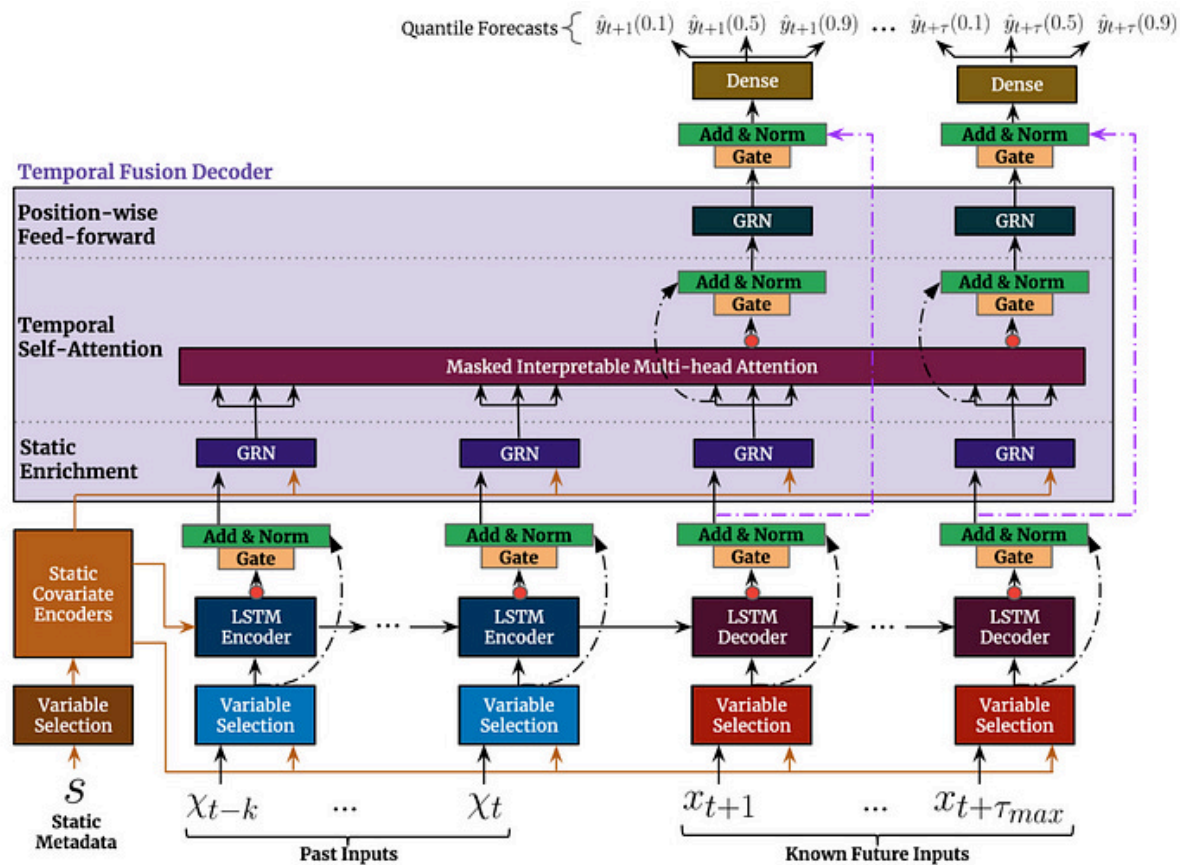


Figure 1: Model Architecture of the TFT (Lim et al., 2021)

Speech Embeddings

Speech Embeddings

Two models chosen for complementary perspectives:

1. **FinBERT**: BERT-based, pre-trained on financial statements (Yang et al., 2020)
2. **FOMC-RoBERTa**: fine-tuned on FOMC data, hawkish/dovish classification (Shah et al., 2023)

Speech Embeddings

Two models chosen for complementary perspectives:

1. **FinBERT**: BERT-based, pre-trained on financial statements (Yang et al., 2020)
2. **FOMC-RoBERTa**: fine-tuned on FOMC data, hawkish/dovish classification (Shah et al., 2023)

Processing: chunk-and-average (512-token windows, 128-token overlap)

- Assumption: information spread across entire speech, not just the beginning

Dimensionality reduction: $768 \Rightarrow X$ dims via Principal Component Analysis or Factor Analysis

- X is treated as hyperparameter

Speech Alignment

Speeches are held at **irregular** and **daily** frequency: how to align with monthly horizon?

Speech Alignment

Speeches are held at **irregular** and **daily** frequency: how to align with monthly horizon?

1. **Mean aggregation**: unweighted mean over rolling window of X months

Speech Alignment

Speeches are held at **irregular** and **daily** frequency: how to align with monthly horizon?

1. **Mean aggregation**: unweighted mean over rolling window of X months
2. **Exponential decay**: $w_t = \exp(-\lambda \cdot d_t)$ where d_t = days before month m
 - More recent speeches receive higher weight
 - λ treated as fixed; speech window X tuned as hyperparameter

Speech Alignment

Speeches are held at **irregular** and **daily** frequency: how to align with monthly horizon?

1. **Mean aggregation**: unweighted mean over rolling window of X months
2. **Exponential decay**: $w_t = \exp(-\lambda \cdot d_t)$ where d_t = days before month m
 - More recent speeches receive higher weight
 - λ treated as fixed; speech window X tuned as hyperparameter
3. **Attention-based**: key = embedding (after dimensionality reduction), query = mean macro state
 - Given current macro conditions, which past speeches are most informative?
 - Fitted on training data only $\Rightarrow$ no leakage!

Speech Alignment

Speeches are held at **irregular** and **daily** frequency: how to align with monthly horizon?

1. **Mean aggregation**: unweighted mean over rolling window of X months
2. **Exponential decay**: $w_t = \exp(-\lambda \cdot d_t)$ where d_t = days before month m
 - More recent speeches receive higher weight
 - λ treated as fixed; speech window X tuned as hyperparameter
3. **Attention-based**: key = embedding (after dimensionality reduction), query = mean macro state
 - Given current macro conditions, which past speeches are most informative?
 - Fitted on training data only $\Rightarrow$ no leakage!

All methods also track: voter speech ratio, chair speech ratio, gender share

Hyperparameter Tuning

Tune in two stages:

Hyperparameter Tuning

Tune in two stages:

1. **Stage 1: Architecture (Macro-Only TFT)**

- Bayesian search via Optuna (TPE sampler)
- 2-fold CV, up to 50 trials
- Includes: encoder length, hidden size, normalizer

Hyperparameter Tuning

Tune in two stages:

1. **Stage 1: Architecture (Macro-Only TFT)**

- Bayesian search via Optuna (TPE sampler)
- 2-fold CV, up to 50 trials
- Includes: encoder length, hidden size, normalizer

2. **Stage 2: Speech Embedding Params**

- Architecture fixed from Stage 1
- Tune: aggregation, reduction, PCA dims, speech window

Evaluation Protocol

- **Walk-forward cross-validation:** expanding window, never use future data

Evaluation Protocol

- **Walk-forward cross-validation:** expanding window, never use future data
- **Metric:** MAE (mean absolute error), RMSE, and relative RMSE

Evaluation Protocol

- **Walk-forward cross-validation:** expanding window, never use future data
- **Metric:** MAE (mean absolute error), RMSE, and relative RMSE
- **Multi-step forecasting:** predictions in non-overlapping 12-month windows
 - Context: training data + model's own previous predictions
 - Same assumption for AR, ARIMA and TFT $\Rightarrow$ fair comparison

Evaluation Protocol

- **Walk-forward cross-validation:** expanding window, never use future data
- **Metric:** MAE (mean absolute error), RMSE, and relative RMSE
- **Multi-step forecasting:** predictions in non-overlapping 12-month windows
 - Context: training data + model's own previous predictions
 - Same assumption for AR, ARIMA and TFT $\Rightarrow$ fair comparison

$\Rightarrow$ All models evaluated on the same information set

Results

Holdout Evaluation

Holdout Metrics: All Models & Horizons

h	Model	CPI		GDP		UNRATE	
		MAE ↓	RMSE ↓	MAE ↓	RMSE ↓	MAE ↓	RMSE ↓
3	AR	0.00218	0.00252	0.005	0.00511	0.24238	0.3012
	ARIMA	0.00122	0.00146	0.005	0.00511	0.25415	0.31466
	TFT	0.00209	0.00239	0.00324	0.00343	0.44346	0.49451
	TFT+Emb	0.00245	0.00276	0.00301	0.00324	0.55122	0.57663
6	AR	0.00218	0.00252	0.005	0.00511	0.24238	0.3012
	ARIMA	0.00122	0.00146	0.005	0.00511	0.25415	0.31466
	TFT	0.00336	0.00363	0.00327	0.00345	0.82816	0.86254
	TFT+Emb	0.00271	0.00296	0.00271	0.00298	0.86496	0.9042
12	AR	0.00218	0.00252	0.005	0.00511	0.24238	0.3012
	ARIMA	0.00122	0.00146	0.005	0.00511	0.25415	0.31466
	TFT	0.00311	0.00339	0.00133	0.00172	3.18856	3.27634
	TFT+Emb	0.00546	0.00559	0.00112	0.00155	0.83562	0.92786

Answering Our Research Question

CPI				GDP			UNRATE		
h	RMSE _{TFT} ↓	RMSE _{+Emb} ↓	relRMSE ↓	RMSE _{TFT} ↓	RMSE _{+Emb} ↓	relRMSE ↓	RMSE _{TFT} ↓	RMSE _{+Emb} ↓	relRMSE ↓
3	0.00239	0.00276	1.1548	0.00343	0.00324	0.9446	0.49451	0.57663	1.1661
6	0.00363	0.00296	0.8154	0.00345	0.00298	0.8638	0.86254	0.9042	1.0483
12	0.00339	0.00559	1.6490	0.00172	0.00155	0.9012	3.27634	0.92786	0.2832

Holdout Predictions ($h=12$)

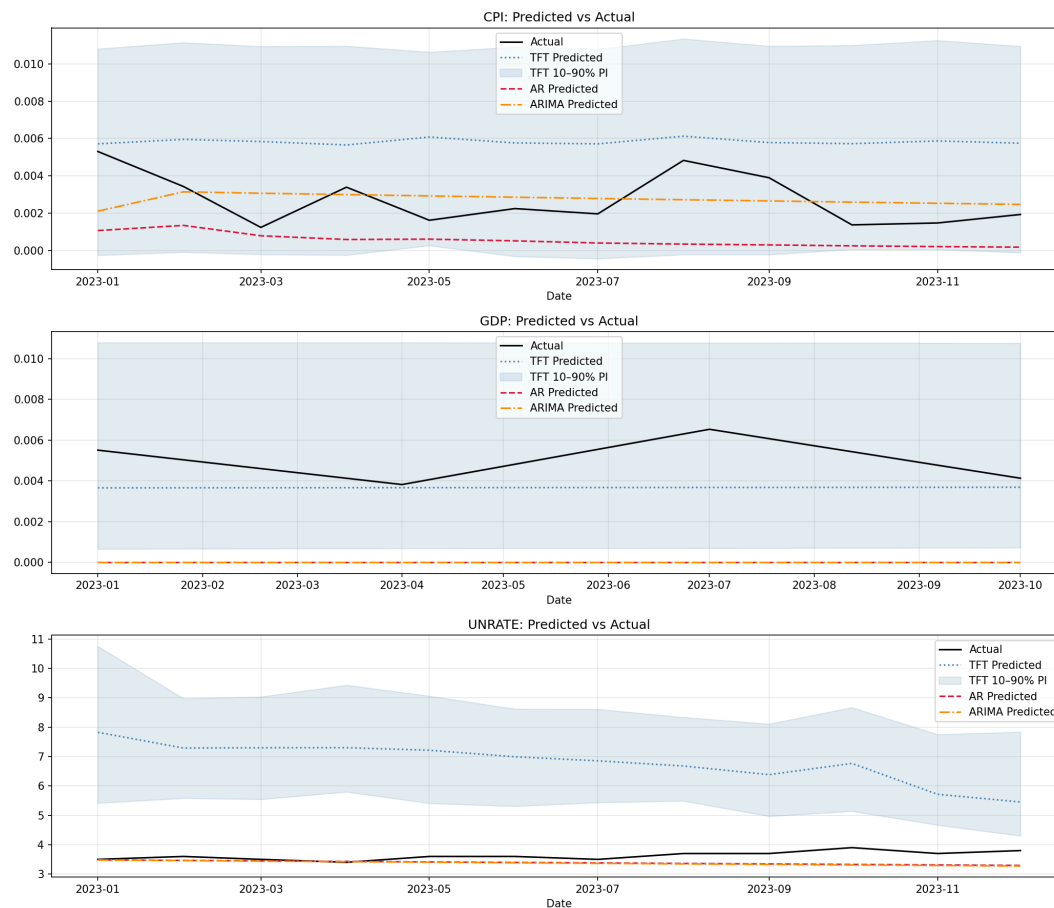


Figure 2: Holdout Predictions $h=12$, Macro Only

Holdout Predictions ($h=12$)

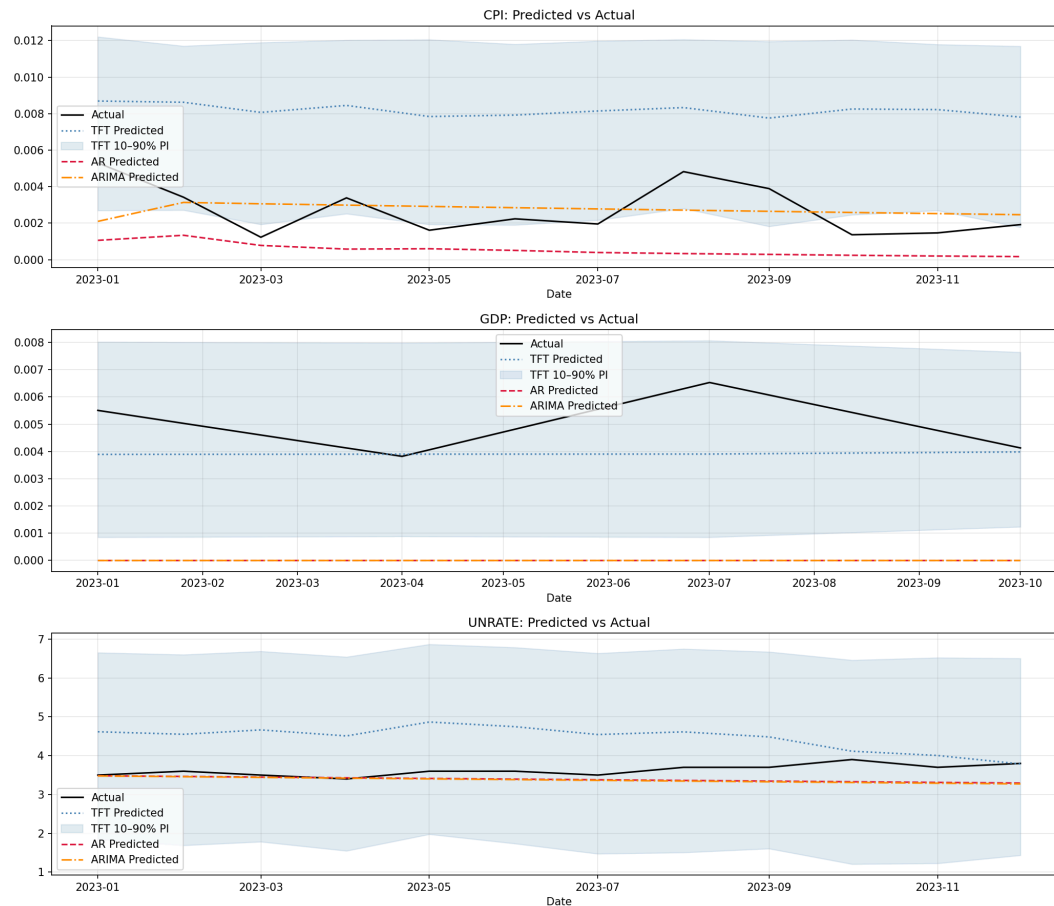


Figure 3: Holdout Predictions $h=12$, Including Embeddings

Holdout Predictions ($h=6$)

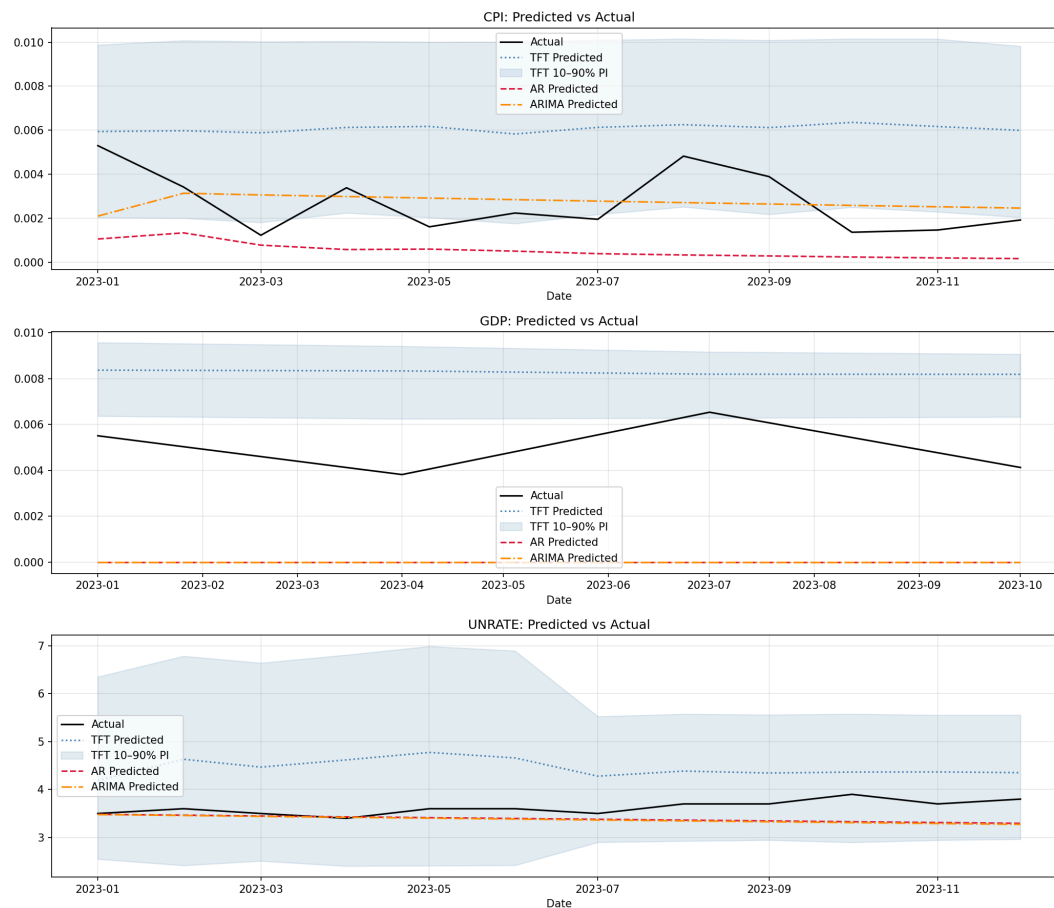


Figure 4: Holdout Predictions $h=6$, Macro Only

Holdout Predictions ($h=6$)

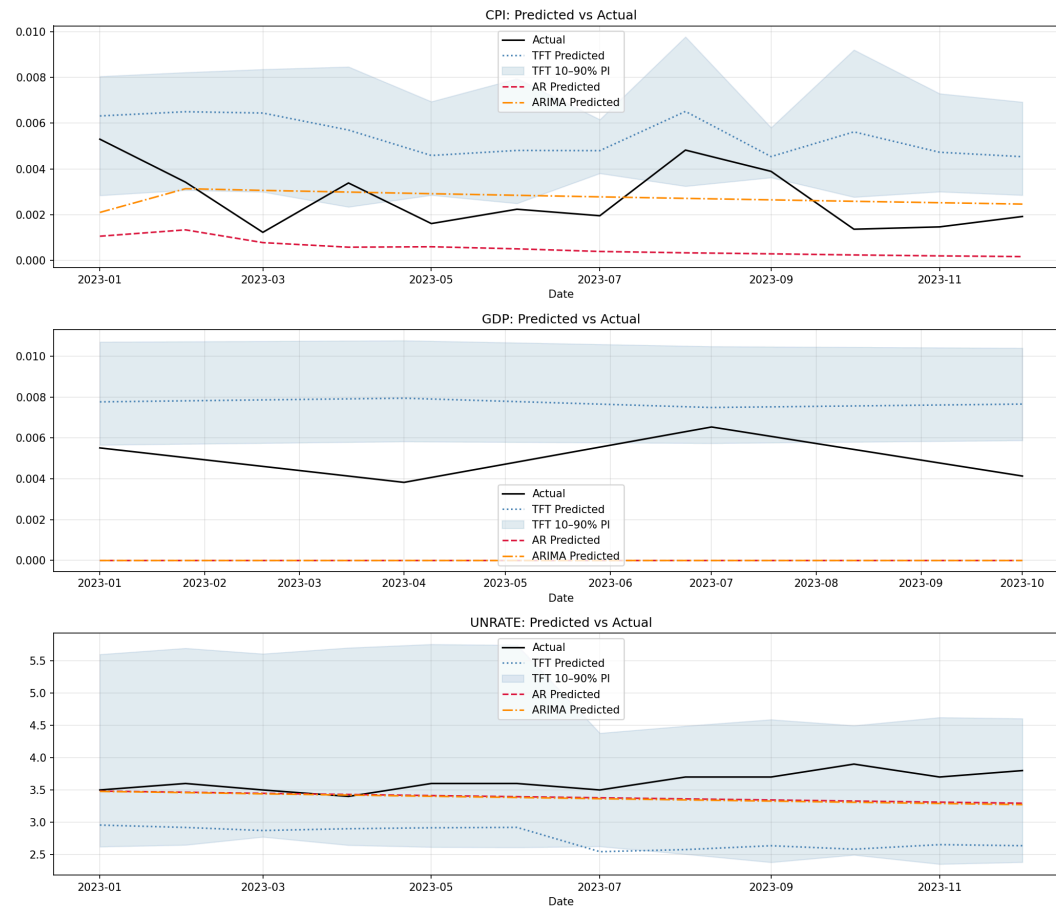


Figure 5: Holdout Predictions $h=6$, Including Embeddings

Holdout Predictions ($h=3$)

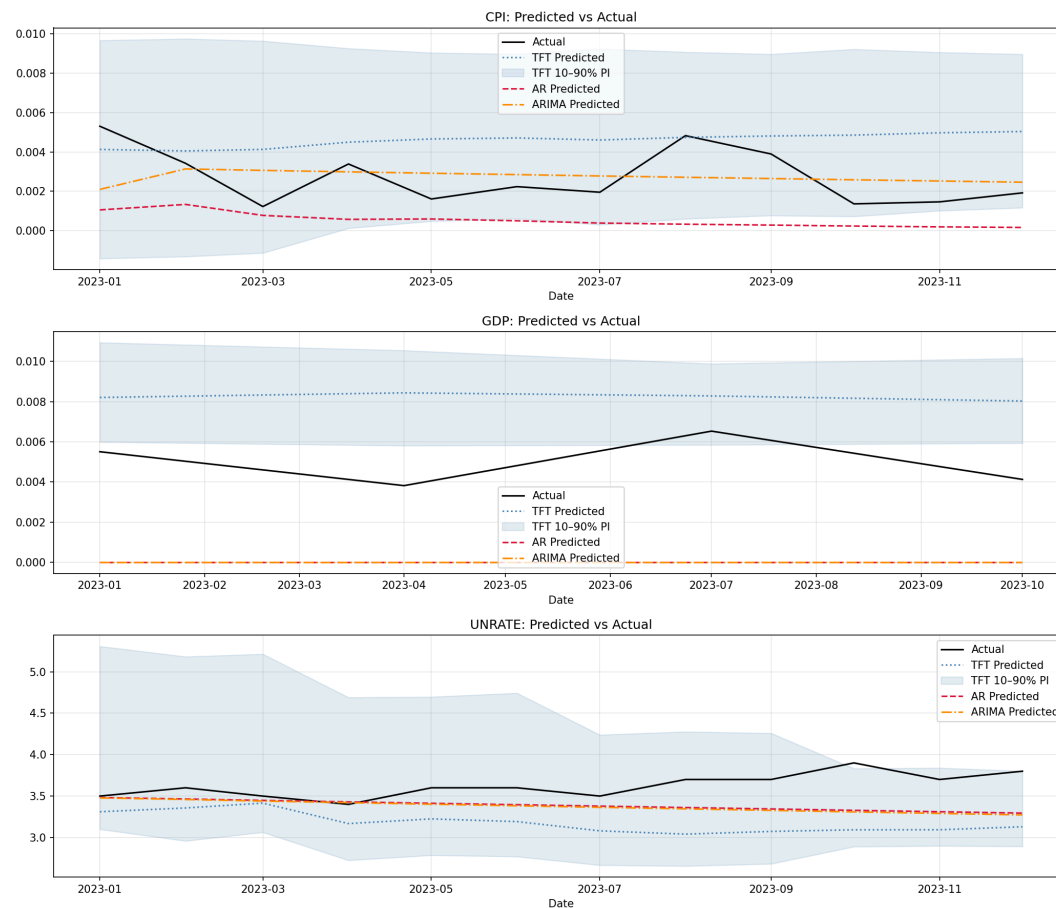


Figure 6: Holdout Predictions $h=3$, Macro Only

Holdout Predictions ($h=3$)

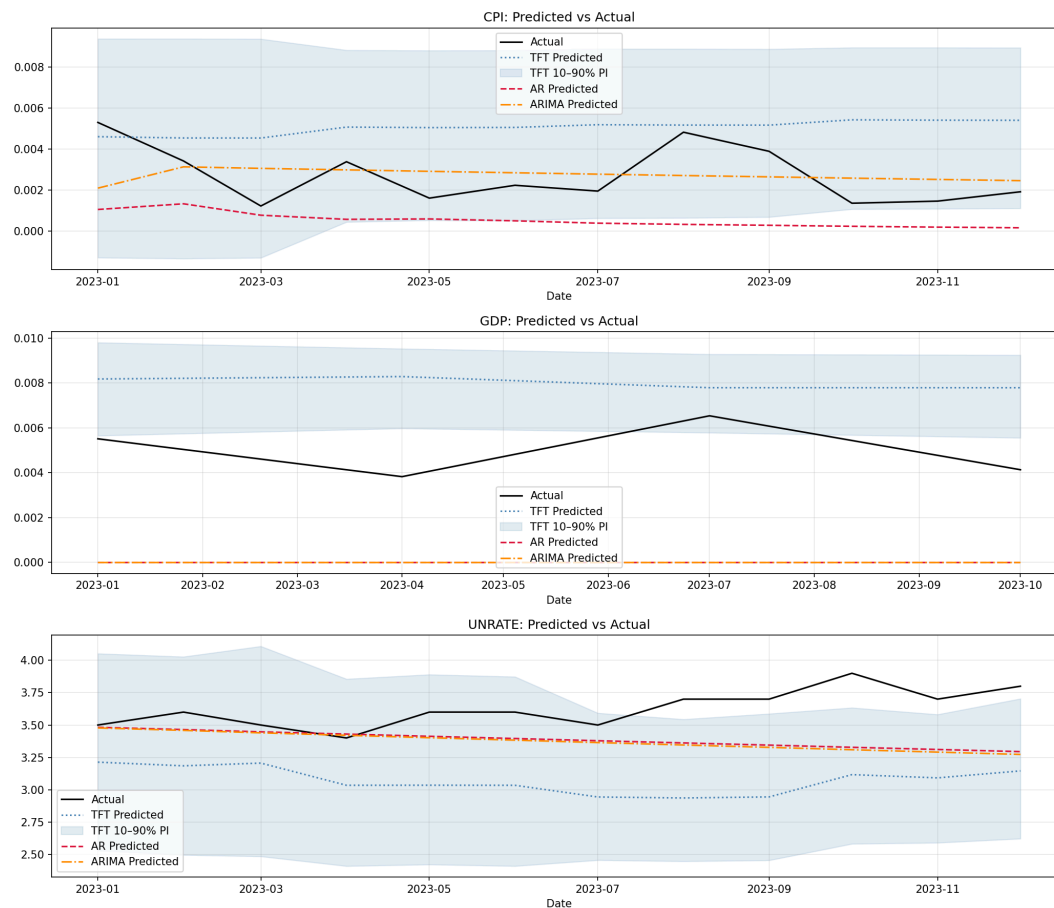


Figure 7: Holdout Predictions $h=3$, Including Embeddings

Robustness Checks

- Shuffling Embeddings
- Using unrelated text (German Texts of Franz Kafka)
- Use not full speech but only first 512 tokens of speech???

Future Work

- Hyperparameter tuning: switch to one stage tuning!
 - context-dependent attention
 - only FOMC-speeches
- Explore New Methods:
 - Use Foundational Model (i.e. TabPFN, TimesFN)
 - Try out SSMs for forecasting
 - Combine with TFT: replace LSTM with SSM for sequence encoder
- Add Vintages: use real-time macro data (as available at forecast time) for more realistic evaluation
- Extend dataset: Working Paper (Lustenberger et al., 2026)
 - 10,000+ speeches since 1914, continuously updated

Conclusion

- in terms of parameters: TFT >> ARIMA > AR
- TFT naturally includes uncertainty bounds
- currently speeches mostly add Noise for 2/3 targets

Bibliography

- Lim, B., Arık, S. Ö., Loeff, N., & Pfister, T. (2021). Temporal Fusion Transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4), 1748–1764. <https://doi.org/https://doi.org/10.1016/j.ijforecast.2021.03.012>
- Lustenberger, T., Rossi, E., & Zeitz, A. (2026). *Central Bank Communication: New Data and Stylized Facts From a Century of Fed Speeches* (Working Paper No. 6/2026). <https://doi.org/tbd>
- Shah, A., Paturi, S., & Chava, S. (2023,). Trillion Dollar Words: A New Financial Dataset, Task & Market Analysis. *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4447632
- Yang, Y., UY, M. C. S., & Huang, A. (2020,). *FinBERT: A Pretrained Language Model for Financial Communications*. <https://arxiv.org/abs/2006.08097>

Appendix

Variable List: Macroeconomic Variables

1. Target variables

- CPI, UNRATE, GDP (quarterly: forward-fill missing months) $\Rightarrow$ AR and ARIMA only take targets

2. Covariates

- Monthly variables: payroll (PAYEMS), industrial production (INDPRO), hours worked in manufacturing (AWHMAN), OECD composite leading indicator (USACLI)
- Weekly / daily variables: exchange rates (GBP, YEN), effective federal funds rate (FFR), Fed balance sheet (WALCL)
 - Higher-frequency variables: take mean **and** std per month (volatility)
- Lags at 1, 2, 6, 12 months for all target variables

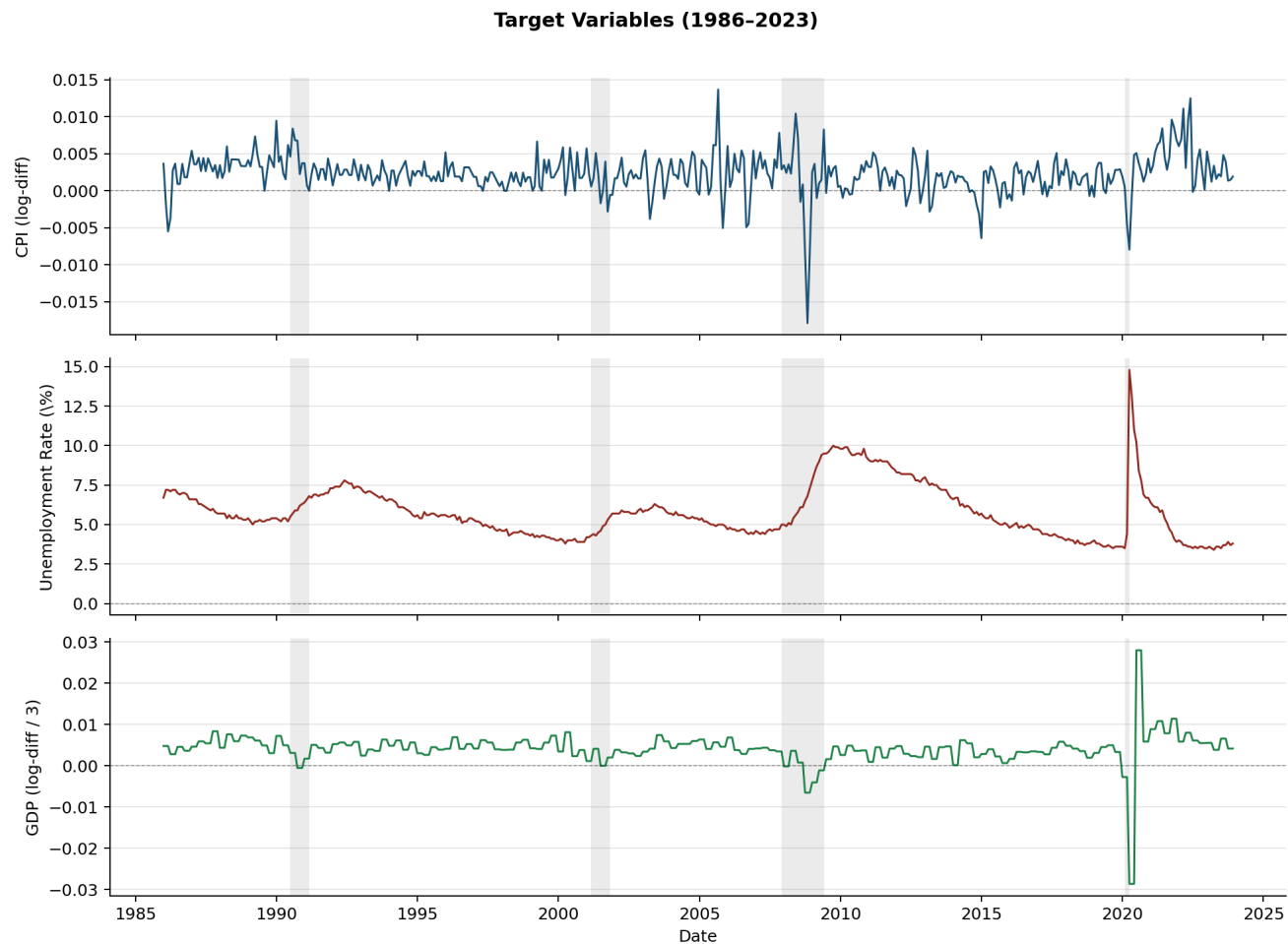
3. Preprocessing: log-difference of CPI, GDP, PAYEMS, INDPRO, AWHMAN

Data Frame:

date	CPI	PAYEMS	INDPRO	UNRATE	AWHMAN	USACLI	GDP	GBP_mean	GBP_std	YEN_mean	YEN_std	FFR_mean
1986-01	0.0036	0.0012	0.0055	6.7	-0.0049	0.0002	0.0048	1.4244	0.0227	199.8905	3.6638	8.1448
1986-02	-0.0018	0.0012	-0.007	7.2	0.0	0.0001	0.0048	1.4297	0.0352	184.8516	4.586	7.86
1986-03	-0.0055	0.0009	-0.0071	7.2	0.0	-0.0001	0.0048	1.4674	0.0154	178.6938	1.8252	7.479
1986-04	-0.0037	0.0019	0.0014	7.1	-0.0049	-0.0001	0.0028	1.4985	0.0363	175.0918	5.1994	6.988
1986-05	0.0028	0.0013	0.0015	7.2	0.0049	-0.0002	0.0028	1.5211	0.021	167.0314	3.4488	6.8529
1986-06	0.0037	-0.0009	-0.004	7.2	0.0	-0.0003	0.0028	1.5085	0.0157	167.5419	2.5161	6.9167

FFR_std	T10Y2Y_mean	T10Y2Y_std	dissent_rate_diss	dissent_net_hawk	dissent_net_hawk_s	n_tighter_sum	n_easier_sum	many_dissent_rec	days_since_fomc	days_to_fomc	fomc_cycle_pos	meeting_this_mon
1.0103	1.0514	0.0368	0.0922	0.25	2.0	5.0	3.0	1.0	15.0	42.0	0.2632	0.0
0.1221	0.7368	0.2171	0.0922	0.25	2.0	5.0	3.0	1.0	46.0	11.0	0.807	1.0
0.2745	0.564	0.0644	0.1131	0.0	0.0	5.0	5.0	1.0	17.0	31.0	0.3542	0.0
0.2155	0.6041	0.0734	0.1292	0.0	0.0	5.0	5.0	1.0	0.0	49.0	0.0	1.0
0.085	0.6424	0.0375	0.1131	0.0	0.0	5.0	5.0	1.0	30.0	19.0	0.6122	1.0
0.175	0.6148	0.1007	0.114	0.0	0.0	5.0	5.0	1.0	12.0	38.0	0.24	0.0

Target Vars

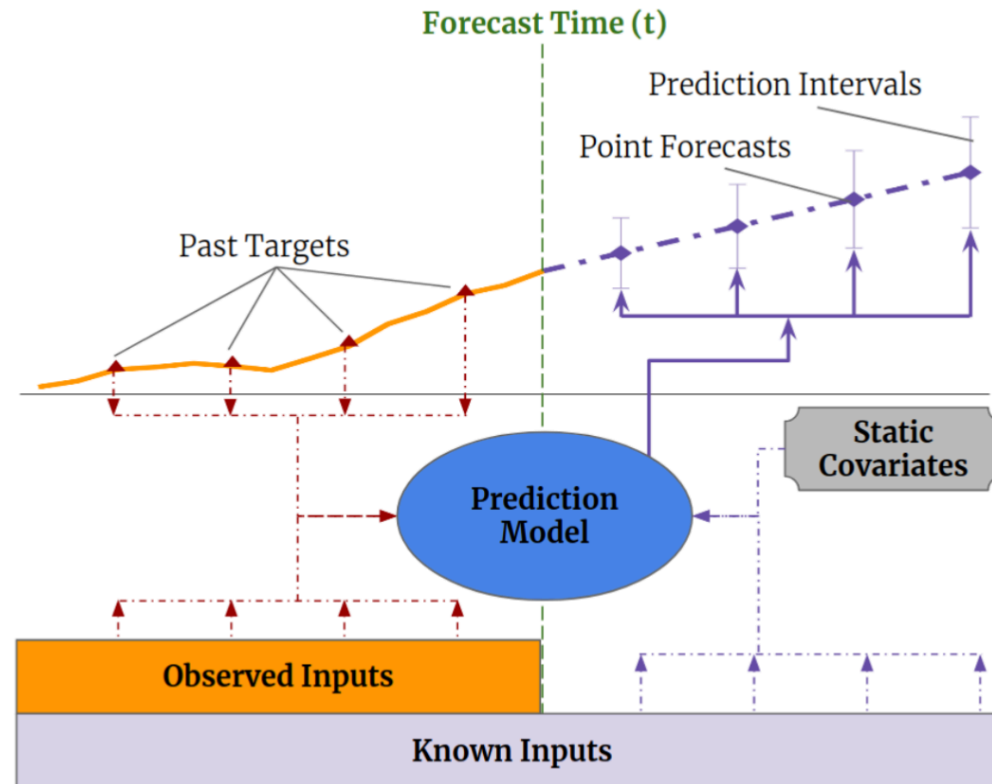


Variable List

The TFT takes four different types of variables:

1. **Static categoricals:** time-invariant, categorical
 - Identifier and frequency of each time series
2. **Static reals:** time-invariant, continuous
 - Number downloads time series from FRED, how long time series has existed
3. **Time-Varying Known Reals:** time-varying, known
 - Day of week, month, year; if day is holiday
 - FOMC Meeting dates
4. **Time-Varying Unknown Reals:** time-varying, unknown
 - Macroeconomic variables, speech embeddings, FOMC dissents

TFT: Input Types



Dimensionality Reduction

FinBERT and FOMC-RoBERTa embeddings are of dimension 768 per speech

⇒ Dimensionality reduction needed for TFT input

We test two methods (number of components X is hyperparameter, fit only on training data):

1. **PCA**: projects embeddings onto X principal components
 - 20 components $\approx$ 80% of variance explained
2. **Factor Analysis (FA)**: models embeddings as linear combination of X latent factors
 - Unlike PCA: explicitly models noise, focuses on **shared** variance
 - Parameters estimated via EM algorithm

Hyperparameter Tuning: Stage 1 Search Space

Architecture tuned on macro-only TFT (per target $\times$ horizon):

Parameter	Type	Range / Values	Nr. of Trials
Encoder length	int	12 – 48	20
Hidden size	categorical	{8, 16, 32, 64, 128, 256}	20
Hidden continuous size	categorical	{2, 4, 8, 16}	20
Dropout	float	0.05 – 0.55	20
Learning rate	log-uniform	10^{-4} – 0.15	20
Normalizer	categorical	{encoder-none, group}	20

Fixed: `lstm_layers = 2, attention_head_size = 2,`

`max_epochs = 50, patience = 15, batch_size = 128`

Hyperparameter Tuning: Stage 2 Search Space

Speech embedding params tuned (architecture fixed from Stage 1):

Parameter	Type	Range / Values
Aggregation	categorical	{mean, decay, attention}
Reduction	categorical	{pca, fa}
PCA components (X)	int	5 – 30
Speech window (months)	int	3 – 12

Tuned separately for each of 9 combinations: {CPI, GDP, UNRATE} $\times$ {3, 6, 12} months

Hyperparameter Tuning: Stage 2 – Optimal Hyperparameters

	FOMC-RoBERTa			FinBERT		
Target	$h=3$	$h=6$	$h=12$	$h=3$	$h=6$	$h=12$
CPI	Mean PCA, $n = 21$ $W = 3$ MAE = 0.001847	Attention FA, $n = 10$ $W = 8$ MAE = 0.001849	Attention PCA, $n = 12$ $W = 9$ MAE = 0.001842	Attention FA, $n = 20$ $W = 5$ MAE = 0.001846	Decay PCA, $n = 23$ $W = 3$ MAE = 0.001846	Mean PCA, $n = 8$ $W = 7$ MAE = 0.001847
GDP	Mean PCA, $n = 8$ $W = 7$ MAE = 0.00154	Attention FA, $n = 12$ $W = 8$ MAE = 0.00171	Attention FA, $n = 5$ $W = 10$ MAE = 0.00152	Attention FA, $n = 6$ $W = 12$ MAE = 0.00170	Attention FA, $n = 10$ $W = 8$ MAE = 0.00157	Decay PCA, $n = 20$ $W = 12$ MAE = 0.00151
UNRATE	Decay FA, $n = 12$ $W = 11$ MAE = 1.31293	Attention FA, $n = 6$ $W = 12$ MAE = 1.31501	Mean PCA, $n = 8$ $W = 7$ MAE = 1.11597	Mean FA, $n = 23$ $W = 5$ MAE = 1.26152	Mean PCA, $n = 14$ $W = 4$ MAE = 1.30896	Mean PCA, $n = 25$ $W = 4$ MAE = 1.20808

where W refers to the speech window size and n to the optimal number of components for the dimensionality reduction. The best result per target×horizon is shown in bold, and h stands for forecast horizon.